

Perez Power Budget

Team Number:	308
Project Name:	
Team Member Names:	Adrian, Jacob, Sam B, Sam M, Andrew, Mo
Version:	2

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,							
All Major Components	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	ESP32	ESP32-S3-WROC	+3 - 3V	1	1000	1000	mA
	DC Motor	PPN7PA12C1	5	1	567	567	mA
					1567	1567	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in							
+12V Power Rail	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	3.3V Regulator	A4402KLPTR-T	1.8V- 3.3V	1	250	250	mA
	5V Regulator	A4402KLPTR-T	3.3V - 5V	1	1000	1000	mA
						0	mA
						0	mA
						0	mA
						0	mA
						1250	mA
						25%	
						1562.5	mA
c1. Regulator or Source Ch	+12V regulator	LM7812	+12V - 35V	1	6000	6000	mA
						4437.5	mA
+5V Power Rail	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	DC Motor	PPN7PA12C1	5	1	567	567	mA
						0	mA
						0	mA
						0	mA
						0	mA
						0	mA
						567	mA
						25%	
						708.75	mA
c2. Regulator or Source Ch	5V Regulator	A4402KLPTR-T	3.3V - 5V	1	2500	2500	mA
						1791.25	mA
+3.3V Power Rail	Component Name	Part Number	Voltage Range	#	Maximum Current	Current(mA)	Unit
	ESP32	ESP32-S3-WROC	+3 - 3V	1	1000	1000	mA
						0	mA
						0	mA
						0	mA
						1000	mA
						25%	
						1250	mA

c4. Regulator or Source Ch	3.3V Regulator	A4402KLPTR-T	1.8V- 3.3V	1	2000	2000	mA	
	Total Remaining Current Available on 3.3V Rail					750	mA	
C. For each power rail above, select a specific voltage regulator using the same process as for major component								
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of								
External Power Source 1	Component Name	Part Number	yVoltage	Ran	Outpu	aximumCurre	alCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	full part number	110VAC	+24V	5000	5000	5000	mA
Power Rails Connected to External Power Source 1	+12V regulator	LM7812	+12V - 35V	1	1000	1000	1000	mA
	5V Regulator	A4402KLPTR-T	3.3V - 5V	1	1000	1000	1000	mA
	3.3V regulator	A4402KLPTR-T	1.8V -3.3V	1	250	250	250	mA
	Total Remaining Current Available on External Power Source 1					2750	2750	mA

